Aim

To perform load testing on a Machine Learning API that predicts telecom tower hardware failures and analyze its performance under multiple concurrent users.

Case Study

A telecom company has deployed an AI model that predicts whether a mobile tower is likely to fail.

Thousands of IoT sensors send requests every second.

The company wants to know:

Can the server handle 100 users simultaneously?

What happens if 1,000 users send requests?

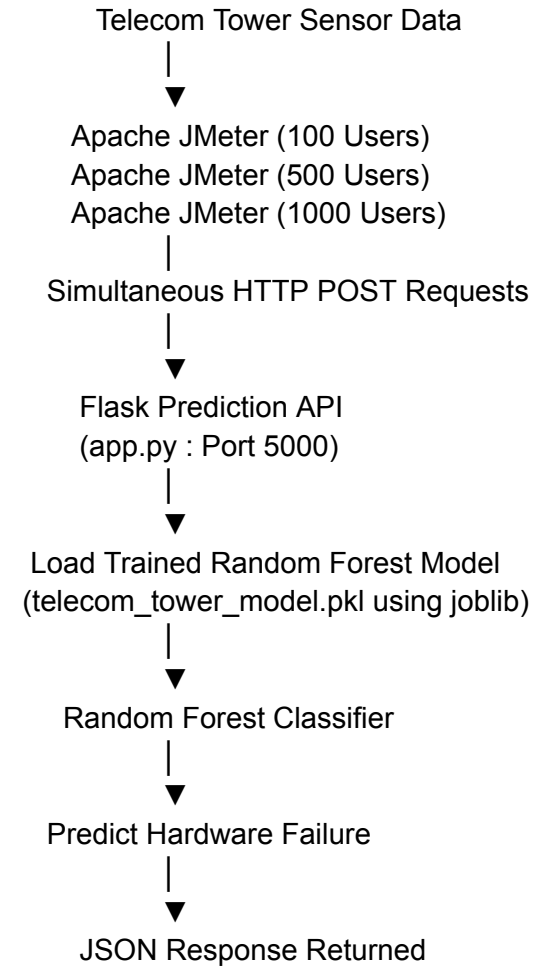
Does the response become slow?

Does the server crash?

This is where Load Testing is performed.

Software Required

Python
Flask
JMeter (Apache JMeter)
Trained Random Forest Model (telecom_tower_model.pkl)



{"prediction":0} or {"prediction":1}

Step 1 – Install Flask

Open Command Prompt

pip install flask

Step 2 – Create Prediction API

Create

app.py

Write

```
from flask import Flask, request, jsonify
import pandas as pd
import joblib

app = Flask(__name__)

model = joblib.load("telecom_tower_model.pkl")

@app.route("/")
def home():
    return "Telecom Tower Prediction API is Running Successfully!"

@app.route("/predict", methods=["POST"])
def predict():

    data = request.get_json()

    df = pd.DataFrame([data])

    prediction = model.predict(df)

    return jsonify({
        "prediction": int(prediction[0])
```

```
})

app.run(debug=True)
```

Step 4 – Test API

Open browser

http://127.0.0.1:5000

Step 5 – Install Apache JMeter

Download Apache JMeter from its official website and extract the ZIP file. Launch jmeter.bat (Windows).

Step 6 – Create Test Plan

Open JMeter

Test Plan
↓
Thread Group

Configure

Property	Value
Number of Users	100
Ramp-Up Period	10 Seconds
Loop Count	5

Meaning

100 users will gradually access the application over 10 seconds.

Step 7 – Add HTTP Request

Thread Group
↓
Add
↓
Sampler
↓
HTTP Request

Configure

Field Value

Server localhost
Port 5000
Method GET
Path

Step 8 – Add JSON Data

Body

{
"voltage":11.2,
"temperature_c":55,
"signal_drop_rate":0.52,
"cpu_usage":85,

"battery_health":62,
"network_latency_ms":90
}

Step 9 – Add Listener

Thread Group
↓
Add
↓
Listener
↓
Summary Report

Also add

View Results Tree

Step 10 – Run Load Test

Click

Start

JMeter starts sending requests.

Expected Output

You will see

Metric Example

Samples 500
Average Response Time 120 ms
Min 60 ms
Max 240 ms
Throughput 85 Requests/sec
Error % 0%

Understanding the Results

Average Response Time

Average time required for one prediction.

Example

120 milliseconds

Lower is better.

Throughput

Number of requests processed every second.

Example

85 Requests/Second

Higher is better.

Error %

Shows failed requests.

0%

means all requests succeeded.

Maximum Response Time

Highest response time observed.

Example

240 ms
